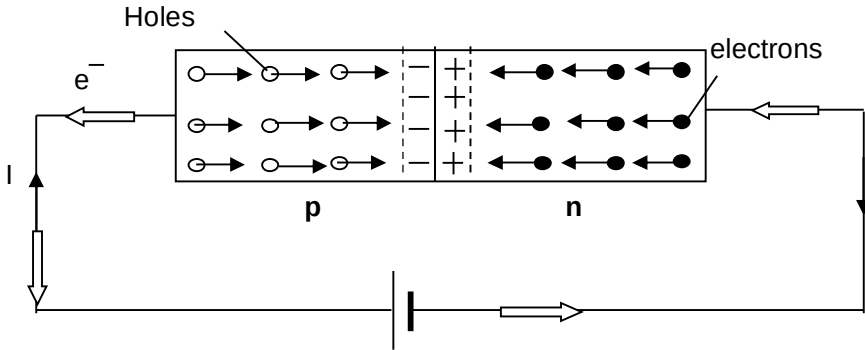


	If rope B is parallel to C, there will not be a horizontal force available and thus equilibrium will not be achieved.
2(a)	P type semiconductor is doped with Group 3 impurities with 1 less valence electron than group 4 silicon. This doping will introduce an acceptor level directly above valence band. The band gap between the acceptor level and the valence band is much narrower than the band gap between the original conduction band and valence band. Thus electrons will be able to excite to acceptor level with less energy and at lower temperature. Holes will be left behind in the valence band that constitute part of the mobile charge carriers. Conductivity of the material will improve with more mobile charge carriers.
2(b)	In p – type semi-conductor the majority carriers are holes, whereas in n – type the majority carriers are electrons. If a battery is now connected to a p – n junction with its positive terminal connected to p – type and negative to n – type, as shown in Fig. 1, then the holes in the p – type tend to move towards the negative and the electrons in n – type towards the positive. Thus a continuous movement of the electrons and holes takes place at the junction and lasts as long as the external battery remains in the circuit. Such a circuit is said to have a forward bias.  Fig. 1 – Forward Bias

If now the battery terminals are reversed, the diode is said to be **reverse biased**. This is illustrated in Fig. 2. The holes in the p end are attracted to the battery's negative terminal and the electrons in the n end are attracted to the positive terminal. The current carriers do not meet near the junction and, ideally, no current flows.

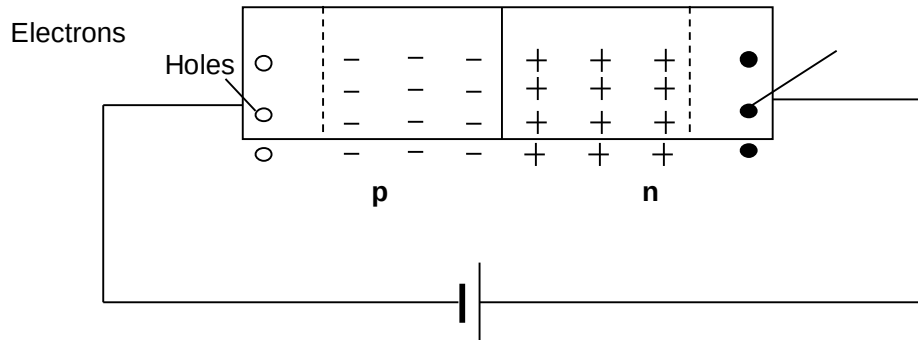


Fig. 2 – Reverse Bias

A current-voltage plot for a p – n junction, showing that the junction is highly conducting when forward-biased and essentially non-conducting in the reverse-biased.

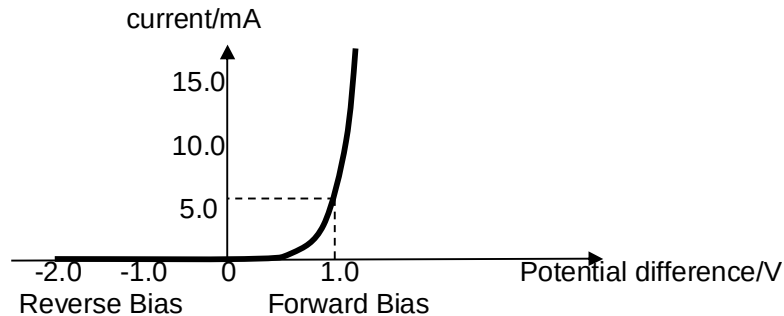


Fig. 3 – Current – Potential Difference Graph

3(a)

The terminal pd across the battery will increase. This is because total resistance